Significance section of an NIH grant proposal

a guide for developing a comprehensive, story-telling first draft

About this document

Terminology: The purpose of any grant-funded project is to solve a problem that has evaded solution or fill an important gap in knowledge, a gap that is preventing the field from moving forward. For some types of projects, the wording 'solve a problem' is fitting. For other types of projects, 'fill a gap in scientific or technical knowledge', relieve a bottleneck', 'o break down a barrier to progress' or 'develop a tool' work better. Below, for brevity, I mostly use 'solve a problem'.

Steps: Not all steps below are relevant for all types of studies; and information from some steps may be redundant with information from other steps. Best approach is to respond to each prompt (to achieve comprehensiveness), then condense.

Headings: You can create whatever headings you think will best guide your reviewer. Try declarative sentences that spoon feed the reader your main points. One point per heading. Bold these.

Bolding, italics and underlining: Permitted. Use judiciously.

References: Citing references shows that you are aware of the gaps in knowledge, roadblocks to progress, opportunities and research underway. You certainly need more than 10 references, but not as many as 100 (this is not a literature review). The sample applications made available by NIH cite ~30-50 references in the Significance section.

To use this guide: Jot down your responses to each of the items below; if an item doesn't seem to apply, skip it. Note that the NIH does not require that you follow their instructions in order, but by doing so you can generate a first draft that is comprehensive and tells a story.

NIH Instruction #1

Explain the importance of the problem or critical barrier to progress in the field that the proposed project addresses.

1. What is the problem / critical barrier to progress / gap in scientific or technical knowledge that the project addresses?
    - State it (this ''problem' will be the central challenge in your field, in the next section you will narrow down to the gap your work will fill; alternatively, if the audience is broad, you may need to start with a big problem, and narrow this down to the central challenge in your field)
    - Describe it / define it / overview it (You might include answers to 'what, when, and where')
    - Document it (give data / statistics / qualitative data) to describe the extent or scope of the problem or barrier
2. Why is this problem/barrier/gap important?
    - State why it is important (for the NIH Institute / for public health / for your field). If the area was identified as a major research focus of an Institute/Center, you should probably cite that.
    - Why should the problem/gap be solved; and why now? If it's not clear from what you have already said, give consequences of not solving problem/filling gap and/or state what progress not solving/filling is preventing
3. What factors are causing or contributing to it? / Why is it occurring? ("*A variety of conditions may ultimately lead to xx… .*" "*The top reason is… .*")
4. Consequences: Who does the problem affect? / What impact is the problem/gap is having on progress? / What will happen (e.g. the symptoms & consequences) if not the problem/gap is not solved/filled?
5. Repeat 1-4 if you are writing for a broad audience and need to start with describing a broad problem

NIH Instruction #2

Describe the scientific premise for the proposed project, including consideration of the strengths and weaknesses of published research or preliminary data crucial to the support of your application.

1. Why does the central challenge continue to be a problem? / Why is a solution currently missing? This will need to narrow down to the problem/gap your work will fill
2. What have people tried?

- Briefly identify the most relevant studies of others. It may be appropriate to identify important, related issues that haven't yet been addressed. For a K award, this might be a good place to explain the relationship between the candidate's research and the mentor's ongoing research program. *("Several promising strategies have been developed to address ___ .")*

3. What part of 'stuff that is needed to solve the central challenge' will your work fill?
    - This is the 'gap' that you are focusing your application around
4. What have others done to try to fill that gap? Why have the other approaches not worked? Were they not the right type, not enough, didn't have whatever resources you/the field now have, failed to appreciate some important factor that is now known ("*However*_____").
5. What has happened that makes you think you and your team have a solution? You will probably describe, briefly, your preliminary data. You will probably have 1-3 headings that state your key findings. You might explain how you came to realize the gap/problem exists. You are allowed to present preliminary data. You may refer to data that you will describe in greater detail in the Approach section.
6. You might end up stating your hypothesis here, since that hypothesis is driven by your preliminary data (and knowledge of the literature).
7. Why is your team qualified to solve the problem/fill the gap? (This might be already clear from the citations of your own published work or references to data figures in the application.)

NIH Instruction #3

Explain how the proposed project will improve scientific knowledge, technical capability, and/or clinical practice in one or more broad fields.

1. What are you doing to do to fill the gap, written generally?
2. In which Aim will you solve which problem?
3. What do you expect to establish? What are your expected outcomes and why you expect them? (e.g., how your solution/approach addresses one or more of the reasons why the problem is happening; how your solution/approach is the best for tackling the problem; how your study will differ from previous work. You can do this Aim by Aim, if that makes sense.)
4. What are the direct benefits to be derived from solving the problem, i.e. scientific benefits, technical abilities benefits, clinical practice benefits, improved diagnostic benefits? Might include how your study differs from previous work and how its results will contribute to the items listed in the instruction.

NIH Instruction #4

Describe how the concepts, methods, technologies, treatments, services, or preventative interventions that drive this field will be changed if the proposed aims are achieved.

1. What will change as a result of your work, i.e. which concepts, methods, technologies, treatments, services, or interventions will change? What is the impact? How will your contribution enable subsequent thinking and research, speed translation and/or improve health (e.g. changes in treatment approaches)? (*"As a result, our knowledge of . . . will be advanced by . . ."*)
2. If not clear from (1), make sure you have stated directly the benefit (direct result), value (broader impact, i.e. how your work will help solve the central challenge), and the potential (longer-term, less-directly-related outcomes); this should convey the exciting-ness of your solution/approach.